Michel Albonico

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Education

2014–2017 **Ph.D.**, *IMT Atlantique Bretagne Pays de la Loire*, Nantes, France
Thesis: Controlling Cloud-Based Systems for Elasticity Testing

2008–2011 **M.Sc.**, *Federal University of Paraná (UFPR)*, Curitiba, Brazil
Thesis: HadoopTest - A Testing Controller for MapReduce-based Systems

2008-2009 **Ed.**, *Federal University of Lavras (UFLA)*, Lavras, Brazil
Monography: A Study of Information Systems Management Applied to the Enterprise SMO Internet Provider

2003–2007 **B.Sc.**, *University of the West of Santa Catarina (UNOESC)*, S. M. Oeste, Brazil
Monography: A Study of the High Availability of a Distributed CORBA System over a VPN

Languages

Portuguese	Native	*Brazilian Portuguese*
English	Advanced	*Long term collaborations with international institutions.*
French	Intermediate-Advanced	*Lived for 3 years in France.*
Spanish	Basic	*Live in the border with Argentina.*
Danish	Learning	

Experience

Professional

2026–Present **Associate Professor**, *SDU*, Vejle, DK
Teaching Software Engineering and Robotics Software Architecture. Researching Sustainable Software Engineering (for Robotics).

2019–Present **Software Engineering Consultant**, *TriXX Software*, Brazil
Mentoring students in cross-platform application development and maintainance.

2022–2025 **Associate Professor**, *UTFPR*, Brazil
Teaching Software Engineering, Robotics Software, and Distributed Programming. Leader of the IntelAgir research group. Permanent member of PPGCA Master's program (since 2024). **Head of the Computer Science section in Francisco Beltrão**.

2012–2021 **Assistant Professor**, *UTFPR*, Brazil
Teaching Software Engineering, Robotics Software, and Distributed Programming.

2018–2020 **Undergraduate Program Coordinator**, *UTFPR*, Brazil
Coordinated a Computer Science program.

2012 **Lecturer and Undergraduate Program Coordinator**, *Unisep*, Dois Vizinhos, Brazil
Coordinated a undergraduate program and taught Computer Science courses.

2011–2012 **Lecturer**, *UNOESC*, S.M Oeste, Brazil
Taught courses in software engineering and programming.

2008–2012 **Technical Course Coordinator and Lecturer**, *FAF*, Barracão, Brazil
Coordinater a technical course and taught foundational computing courses.

2006–2008 **Infrastructure Manager**, *SMO Internet Provider*, S.M. Oeste, Brazil
Managed network infrastructure and ensured service reliability.

2004–2006 **Enterpreneur**, *Dominio Informática*, Dionisio Cerqueira, Brazil
Technical support and software maintainance.

Training

2026 **Danish Course - Module 2**, *Sprogcenter Vejle*, Vejle Municipality
80 hours

2026 **Danish Course - Module 1**, *Sprogcenter Vejle*, Vejle Municipality
80 hours

2025 **Generative AI for Software Development Professional Certificate** , *DeepLearning.AI*, Coursera
15 hours

2024 **ChatGPT Prompt Engineering for Developers**, *DeepLearning.AI*, Coursera
1 hour

2022 **Winter School in Software Engineering**, *National Research University Higher School of Econ, HSE*, Russia
33 hours

2020 **CAA Training for BNI**, *INEP/MEC*, Brazil
24 hours

2016 **TAROT Summer School**, *Télécom Sud Paris*, France
40 hours

2016 **French for Foreigners (Advanced)**, *Maison des Chercheurs Étrangers, MCE*, France
30 hours

2015 **Grammar and Punctuation**, *University of California, Irvine, UCI*, USA
20 hours

2014–2015 **My English Online**, *Cengage Learning / CAPES, MEO*, Brazil
240 hours

2014–2015 **French Course**, *Ecole des Mines de Nantes, EMN*, France
30 hours

2014 **French Course**, *Ecole des Mines de Nantes, EMN*, France
15 hours

2012–2013 **Intermediate and Advanced English**, *Open English, OE*, Brazil
160 hours

2011 **MC1 - Cloud Computing: Elastic and Secure**, *Universidade Federal de Santa Catarina, UFSC*, Brazil
8 hours

2011 **Expanding Horizons**, *FISK English Courses, FISK*, Brazil
40 hours

2010 **Flying High**, *FISK English Courses, FISK*, Brazil
40 hours

2009 **Spreading Wings**, *FISK English Courses, FISK*, Brazil
40 hours

2009 **Aiming at the Sky**, *FISK English Courses, FISK*, Brazil
40 hours

2007 **Asterisk VOIP**, *VOffice, VO*, Brazil
40 hours

2007 **Basic 1**, *FISK English Courses, FISK*, Brazil
40 hours

2006 **Structured Optical and Metallic Cabling**, *SENAC - São Miguel do Oeste, SENAC*, Brazil
40 hours

Research Visits/Stays

2026 **Short Visit**, *PPGCA Master's Program, UTFPR*, Brazil
Sovereignity and governability of AI systems.

2026 **Short Visits**, *ISW, University of Stuttgart*, Germany
AI-driven robotics software.

2020–2021 **Postdoctoral Researcher**, *S2 Group, VU Amsterdam*, the Netherlands
Research on green software engineering for robotic systems, focusing on ROS.

2023–2024 **Visiting Researcher**, *ISW, University of Stuttgart*, Germany
Research on ML4GreenROS: Using machine learning to optimize ROS software efficiency.

2024 **Short-Term Research Visit**, *INESC TEC, University of Porto*, Portugal
Research on static analysis of ROS software for energy efficiency.

Research Interests

Primary Focus: Sustainable Software Engineering, Responsible AI for Software Engineering, Robotics Software Engineering, ROS-based Robotics Software, and Robotics in Education.

Current Collaborations: Prof. Andreas Wortmann, ISW@University of Stuttgart, Germany

Prof. Seonah Lee, Gyeongsang National University, South Korea.

Prof. Ivano Malavolta, S2 Group@VU Amsterdam, the Netherlands.

Prof. Nuno Macedo, INESC/TEC@University of Porto, Portugal

Prof. Felipe Fronchetti, Luisiana State University (LSU), USA
PPGCI Master's Program - UTFPR, Cornélio Procópio, Brazil
PPGCA Master's Program - UTFPR, Curitiba, Brazil

Research Projects

2026–Present **Assess GenAI**, *IT-Vest: AAU, AU, and SDU*
A framework for exploring the disciplinary transformations initiated by generative AI, and how these transformations create new demands for the (digital) competences of our students and graduates.

2026–Present **Sovereignity and Sustainability of Robotics Software in the AI Age**, *SDU, UTFPR*
Qualitative study on sovereign infrastructures for the use of AI in software engineering.

2024–Present **Social Aspects of ROS Projects**, *UTFPR/SDU, LSU*
Qualitative study of social aspects in open-source ROS projects.

2024–2026 **HighConfigurable ROS**, *UTFPR/SDU, University of Porto*
Static analysis of ROS software projects aiming to enhance energy efficiency and performance.

2023–2026 **AI4GreenROS**, *UTFPR/SDU, University of Stuttgart, VU Amsterdam*
Developing AI-based adaptive configurations to improve performance and energy efficiency in ROS-based robotic systems.

2020–2023 **ElasIoT: Elastic Solutions for IoT Software**, *UTFPR*
Investigating scalable and adaptable cloud/edge strategies for IoT environments to enhance system performance and resource utilization.

2020–2023 **GreenROS**, *UTFPR, VU Amsterdam*
Studying energy-efficient software architectures for robotic systems, leveraging software repository mining techniques.

2020–2022 **Green Tactics for Robotics Software**, *VU Amsterdam*
Developing software engineering tactics for optimizing energy consumption in robotic applications.

2016–2017 **Dizolo**, *IMT Atlantique*
Exploring constraints on data in cyber-physical systems for factories of the future.

2011–2011 **TaaS: Testing Applications for Cloud Computing**, *UTFPR, INRIA, UFSC*
Building a framework for large-scale cloud testing using fault injection and test automation.

Research Funding

2026 **Trips to Visit International Partners**, funded by Carslberg Foundation - Denmark (≈ DKK 33,000 / EUR 5,000).

2025-Present **Visiting Expert Researcher (Short-term)**, funded by ICM - Germany (≈ BRL 65,117 / EUR 10,018).

2024-2025 Robotics Software Innovation Project, **Researcher**, funded by SEBRAE - Brazil (BRL 150,000 / ≈ EUR 25,000).

2024 **Visiting Researcher (Short-term)**, funded by INESC TEC, University of Porto - Portugal (≈ BRL 30,000 / EUR 5,000).

2024 ML4GreenROS Project, **Coordinator/Postdoctoral Researcher**, funded by CNPq - Brazil (BRL 141,000 / ≈ EUR 23,500).

2023-2024 **Visiting Expert (Mid-term)**, funded by ICM - Germany (≈ BRL 124,200 / EUR 20,700).

2020-2021 **Postdoctoral Fellow**, funded by NWO - the Netherlands (≈ BRL 165,000 / EUR 27,500).

2018-2019 Smart Devices Project, **Coordinator/Developer**, funded by UTFPR - Brazil (BRL 30,000 / ≈ EUR 5,000).

2018-2020 ElasIoT Project, **Coordinator/Researcher**, funded by CNPq and Fundação Araucária - Brazil (BRL 39,940 / ≈ EUR 6,650).

2017-2018 IoTriXX Dashboard Project, **Coordinator/Researcher/Developer**, funded by Fundação Araucária - Brazil (BRL 120,000 / ≈ EUR 20,000).

2017-2018 SmartLock Project, **Researcher/Developer**, funded by Fundação Araucária - Brazil (BRL 120,000 / ≈ EUR 20,000).

2014-2017 Cloud Elasticity Testing Project, **Coordinator/Ph.D. candidate**, funded by CAPES - Brazil (BRL 690,000 / ≈ EUR 115,000).

Academic Outreach

2023-2025 Transformar Project (building equipment for robotics/programming teaching), **Researcher**, Public Elementary Schools, Extension/Innovation Project, 2 hours/week

2022-2023 Robotics in Education, **Coordinator**, Public Elementary Schools, Extension Project, 30 hours

2019-2021 Safety at Work App, **Coordinator/Tech Leader**, Vibra SA (private company), Innovation Project, 100 hours

2018-2019 Computing and Electronic Prototyping with High-Ability Students, **Coordinator/Teacher**, Public Elementary Schools, Extension Project, 30 hours

Leadership

2025-2025 Head of Francisco Beltrão's Computer Science Department, UTFPR - Brazil.

2018-2025 Head of IntelAgir Research Group, UTFPR - Brazil.

2018-2020 Computer Science undergraduate program Coordinator, UTFPR - Brazil.

2011 Computer Science undergraduate program Coordinator, Unisep - Brazil.

2009-2011 Computer Science undergraduate program Coordinator, FAF - Brazil.

Mentorship

Ongoing Supervisions

Master's Theses

2026-Present Everton Gonçalves. Towards a Green Testing Pipeline for Java Applications. Master's thesis in Informatics - Federal University of Technology - Paraná (UTFPR). (Supervisor).

2026-Present Angemydelson de Saint. Analyzing the Impact of AI in the Energy Efficiency of Android Applications. Master's thesis in Informatics - Federal University of Technology - Paraná (UTFPR). (Supervisor).

2025-Present Walysson Duarte. Analysing the benefits of the Elixir library (RCLex) on programming highly concurrent robotics software. Master's thesis in Informatics - Federal University of Technology - Paraná (UTFPR). (Co-supervisor).

2025-Present Douglas de Souza. Challenges for a donation platform orchestrated over blockchain. Master's thesis in Informatics - Federal University of Technology - Paraná (UTFPR). (Co-supervisor).

2025-Present Rodrigo Iser. Analysis of Performance and Energy Consumption of Simulation Tools for ROS. Master's thesis in Informatics - Federal University of Technology - Paraná (UTFPR). (Co-supervisor).

2024-Present Ricardo de Souza. A Gamification Tool for Teaching Green/Energy-Efficient Programming. Master's thesis in Applied Computing - Federal University of Technology - Paraná (UTFPR). (Co-supervisor).

Completed Supervisions

Master's Theses

2024-2026 Maciel Felipe Borges. A Study on the Evolution of the Robot Operating System (ROS) with a Focus on Energy Efficiency. Master's thesis in Informatics - Federal University of Technology - Paraná (UTFPR). (Co-supervisor).

2024 Alexander Kraus. Continuous Optimization of ROS Configurations for Energy Efficiency. Master's thesis in ISW - University of Stuttgart. (Co-supervisor).

2021 Milica Dordevic. Empirical Evaluation of Computation Offloading Strategies in Robotic Systems. Master's thesis in Computer Science - Vrije Universiteit. Co-supervisor: Michel Albonico.

2021 Kwame Chan Jong Chu. A Study on the Resource Utilization of SLAM Algorithms on Simulated and Real-World ROS Systems. Master's thesis in Computer Science - Vrije Universiteit. Co-supervisor: Michel Albonico.

2021 Engel Hamer. Resource Utilization of 2D SLAM Algorithms in ROS-Based Systems: an Empirical Evaluation. Master's thesis in Computer Science - Vrije Universiteit. Co-supervisor: Michel Albonico.

Bachelor Project

2024-2026 Manuela Bechara Cannizza. Understanding the impact of programming language on ROS energy efficiency. (Information Systems - FB) - Federal University of Technology - Paraná (UTFPR). (Supervisor).

2025 Pedro Peruzzo. Course: Software Modeling and Design. Teaching Assistant (Information Systems - FB) - Federal University of Technology - Paraná (UTFPR) (Supervisor).

2024-2025 Alexia Celina Nielsen. Empirical Experiments on the Energy Consumption of Robotics Algorithms in ROS - Topic-Based/Publish-Subscribe Communication. Undergraduate Research (Information Systems - FB) - Federal University of Technology - Paraná (UTFPR), Fundação Araucária. (Supervisor).

2025-2026 Phillipp Schlüter. Mining critical navigation points from maps. Co-supervision. Supervisor: Andreas Wortmann.

2023 Juliano Soares. Planning of Basic Electronics Content for Introductory Arduino. Specialization Thesis in Informatics - Federal University of Technology - Paraná (UTFPR). Supervisor: Michel Albonico.

2023 Lucas Augusto Peruffo. Development of an Interaction Device for Immersive Education Environments. Undergraduate Thesis in Informatics - Federal University of Technology - Paraná (UTFPR). Supervisor: Michel Albonico.

2023 Everson Borges da Rosa. InterativeEdu: Prototyping an Interactive Floor as a Learning Tool Using Open Technology. Undergraduate Thesis in Informatics - Federal University of Technology - Paraná (UTFPR). Supervisor: Michel Albonico.

2021 Gustavo Henrique B. Slomski. OiArduBot: A Domain-Specific Language for Teaching Robotics with Open Technology. Undergraduate Thesis in Informatics - Federal University of Technology - Paraná (UTFPR). Supervisor: Michel Albonico.

2022 Alexandre Juarez Corneau. Empirical Study of Software for IoT and Robotics Systems. Undergraduate Research (Informatics) - Federal University of Technology - Paraná (UTFPR), Fundação Araucária. Supervisor: Michel Albonico.

2021 Joe Hopkinson. Benchmarking Path Coverage Algorithms in ROS-Based Systems. Undergraduate Thesis in Computer Science - Vrije Universiteit. Supervisor: Michel Albonico.

2021 Etienne Herbeaux. Benchmarking SLAM Algorithms in ROS-Based Systems. Undergraduate Thesis in Computer Science - Vrije Universiteit. Supervisor: Michel Albonico.

2021 Karl Plagemann. Benchmarking the Energy Efficiency of Different ROS Versions and Configurations. Undergraduate Thesis in Computer Science - Vrije Universiteit. Supervisor: Michel Albonico.

2020 Gustavo Slomski. An Optimized Programming Language for Teaching Educational Robotics with Open Technologies. Undergraduate Thesis in Informatics - Federal University of Technology - Paraná (UTFPR). Supervisor: Michel Albonico.

2012 Cleverson Massaroli. Comparison Between Mahou and Relational Database Management Systems. Undergraduate Thesis in Information Systems - União de Ensino do Sudoeste do Paraná. Supervisor: Michel Albonico.

2012 Paulo Vitor Cogo. Project for the Implementation of Optical Fiber for the Margarida Galvan Neighborhood in Dois Vizinhos-PR. Undergraduate Thesis in Information Systems - União de Ensino do Sudoeste do Paraná. Supervisor: Michel Albonico.

2012 Fernando Zotti and Fernando Socolovski. Web System for Fuel Station Users. Undergraduate Thesis in Information Systems - União de Ensino do Sudoeste do Paraná. Supervisor: Michel Albonico.

2012 Henrique Koerich. A Study on Cloud Computing with Application of Software Quality Testing. Undergraduate Thesis in Information Systems - União de Ensino do Sudoeste do Paraná. Supervisor: Michel Albonico.

2011 Andreia Salete Mayer Eberhart. Automation of Residential Equipment Using Java. Undergraduate Thesis in Internet Systems Technology - Faculdade da Fronteira. Supervisor: Michel Albonico.

2011 Diego Delazeri. System for Updating Championship Classification Tables. Undergraduate Thesis in Internet Systems Technology - Faculdade da Fronteira. Supervisor: Michel Albonico.

2011 Lucas Ecco. Measurement of Degradation by Noise in Wired and Wireless Networks. Undergraduate Thesis in Internet Systems Technology - Faculdade da Fronteira. Supervisor: Michel Albonico.

2011 Everton da Silva. System for Assisting Software Acquisition. Undergraduate Thesis in Internet Systems Technology - Faculdade da Fronteira. Supervisor: Michel Albonico.

2011 Alex Roberto Grunwaldt and Edson Renato Drews. System for Access Ticketing and Ad Mining on News Websites Applied to the Click Amerios Portal. Undergraduate Thesis in Internet Systems Technology - Universidade do Oeste de Santa Catarina. Supervisor: Michel Albonico.

2011 Jean Carlos Copati. Call Management System - Help Desk. Undergraduate Thesis in Internet Systems Technology - Universidade do Oeste de Santa Catarina. Supervisor: Michel Albonico.

2010 Douglas Geova Kaktin. Comparison of Similarity Functions for Open Source Database Management Systems. Undergraduate Thesis in Internet Systems Technology - Faculdade da Fronteira. Supervisor: Michel Albonico.

2010 Marcio Rodrigo Backes. Algorithm for Searching and Sorting Log Files Using the MapReduce Framework. Undergraduate Thesis in Internet Systems Technology - Faculdade da Fronteira. Supervisor: Michel Albonico.

2009 Daniely da Silva Loiola. Cedro News - Information Portal for the City of São José do Cedro. Undergraduate Thesis in Informatics Technology - Faculdade da Fronteira. Supervisor: Michel Albonico.

2009 Cleder F. Perufo. TECHNICALL - System for Technical Support Management of Wireless Providers. Undergraduate Thesis in Informatics Technology - Faculdade da Fronteira. Supervisor: Michel Albonico.

Undergrad. Research

2021 Bruno Cardoso. Preliminary Listing of Works on Robotic Internet of Things (IoRT) Software through Systematic Literature Review. Undergraduate Research (Informatics) - Federal University of Technology - Paraná (UTFPR). Supervisor: Michel Albonico.

2021 Berry Chen. Extracting Architectural Components from ROS Executions. Undergraduate Research (Computer Science) - Vrije Universiteit. Supervisor: Michel Albonico.

2021 Leonardo Pilatti. Programming and Conducting Experiments with TurtleBot. Undergraduate Research (Informatics) - Federal University of Technology - Paraná (UTFPR), Fundação Araucária. Supervisor: Michel Albonico.

2021 Wilson Lira. Automation of Scripts for Data Extraction from Google Scholar. Undergraduate Research (Informatics) - Federal University of Technology - Paraná (UTFPR). Supervisor: Michel Albonico.

2020 Bruno Cardoso. Mining Software Repositories of IoRT. Undergraduate Research (Informatics) - Federal University of Technology - Paraná (UTFPR), Fundação Araucária. Supervisor: Michel Albonico.

Other Supervisions

2024 Li Lin. Development of Artificial Intelligence Models for Selecting ROS Configuration Parameters. ISW - University of Stuttgart. Supervisor: Michel Albonico.

2024 Monisha Chandrashekar. Adaptations of an Open-Source Energy Meter for Robotics Experiments. ISW - University of Stuttgart. Supervisor: Michel Albonico.

Awards and Recognition

2026 Best Reviewer of the European Conference on Software Architecture (ECSA).

2025 Selected as Associate Professor for the Centre for Software Technology at the University of Southern Denmark, Vejle.

Qualified for the position of DIAS Fellow in Engineering at the University of Southern Denmark, Odense - Assistant Professor in Software Engineering.

2024 Elected Head of the Computer Science Department at UTFPR - Francisco Beltrão Campus.

Best Paper of Robotics Software Engineering (RoSE) Workshop.

2023 Awarded a postdoctoral fellowship funded by CNPq (Brazil) for research in Germany.

Received the Best Paper Award at the Workshop on Robotics Software Engineering.

Recognized as a Distinguished Reviewer at the Brazilian Symposium on Informatics in Education.

2018 Elected Coordinator of the Undergraduate Program in Computer Science at UTFPR.

2015 Diplôme d'Études en Lange Française - DELF B1.

2014 Awarded a full Ph.D. scholarship in France, funded by CAPES (Brazil).

2012 Ranked first for the Assistant Professor position at UTFPR.

Professional Services

2026 *Guest Editor* of Frontiers Computer Science Journal

Program Committee Chair of 8th International Workshop on Robotics Software Engineering (RoSE'26).

Program Committee Chair of 2nd Workshop on Software Engineering for Functional Programming (SE4FP).

Program committee member of the 24th Mining Software Repositories (MSR) conference.

Program committee member of the 20th International Conference on Empirical Software Engineering and Measurement (ESEM).

Program committee member of European Conference on Software Architecture (ECSA).

Program committee member of Brazilian Symposium on Software Engineering - Insightful Ideas and Emerging Results Track (SBES-IIER)

Program committee member of Brazilian Symposium on Information Systems (SBSI).

Program committee member of IEEE/RSJ International Conference on INTELLIGENT ROBOTS & SYSTEMS (IROS)

Program committee member of IEEE International Conference on Robotics & Automation (ICRA)

External reviewer of Journal of Systems and Software (JSS).

External reviewer of Journal Frontiers – Robotics and Autonomous Systems.

External reviewer of Journal of Information and Software Technology (IST).

External reviewer of Journal of Transactions on Software Engineering (TSE).

2025 *Program committee chair* of 1st Workshop on Software Engineering for Functional Programming (SE4FP).

Program committee member of European Conference on Software Architecture (ECSA).

Program committee member of 5th LATAM School - The Latin American School in Software Engineering Research .

Program committee member of Brazilian Symposium on Software Engineering - Insightful Ideas and Emerging Results Track (SBES-IIER)

External reviewer of Journal of Brazilian Computer Society.

External reviewer of Frontiers in Robotics and AI.

2024 *External reviewer* of Transactions on Software Engineering and Methodology (TOSEM).

External reviewer of Software Quality Journal (SQJ).

Program committee member of Brazilian Symposium on Informatics in Education (SBIE)

Program committee member of Brazilian Symposium on Software Engineering (SBES)

Program committee member of Workshop on Robotics Software Engineering (RoSE)

External reviewer of Frontiers in Robotics and AI.

Program committee member of Mining Software Repositories (MSR).

Program committee member of International Conference on Intelligent Robots and Systems (IROS).

External reviewer of Journal of the Brazilian Computer Society (JBCS).

External reviewer of Internacional Journal of Robotics Research (IJRR)

External reviewer of European Conference on Software Architecture (ECSA) - Post Proceedings.

External reviewer of Applied Science Open Access Journal.

2023 *Program committee member* of Brazilian Symposium on Software Engineering (SBES) -Insightful Ideas and Emerging Results (IIER)

Program committee member ofInternational Symposium on Software Testing and Analysis (ISSTA) - Artifact Evaluation (AE)

External reviewer of Open Research Europe.

Program committee member of Workshop on Informatics in Schools (WIE), Brazil

Program committee member of Mining Software Repositories (MSR) - Industrial Track

Program committee member of Context-aware, Autonomous, and Smart Architectures (CASA)@ECSA.

2022 *Program committee member* of Workshop on Robotics Software Engineering (RoSE)

Program committee member of International Conference on Open Source Systems (OSS)

Program committee member of Workshop on Mining Software Repositories for Software Architecture (MSR4SA)

Program committee member of Mining Software Repositories (MSR)

Program committee member of Information and Software Technology (IST)

Sub-reviewer of International Conference on Software Engineering (ICSE).

External reviewer of European Conference on Software Architecture (ECSA) - Post Proceedings.

Program committee member of Context-aware, Autonomous, and Smart Architectures (CASA)@ECSA.

2021 *Program committee member* of Workshop on Robotics Software Engineering

External reviewer of Latin American Congress on Free Software and Open Technologies (Latinoware)

External reviewer of Journal of the Brazilian Computer Society (JBCS).

External reviewer of IEEE Software.

External reviewer of IEEE Sensors Letters.

Sub-reviewer of International Conference on Software Engineering (ICSE).

Program committee member of Context-aware, Autonomous, and Smart Architectures (CASA)@ECSA.

External reviewer of IOP e-books.

2018 *Program committee member* of Regional School of Software Engineering (ERES) - Dois Vizinhos, Brazil

Teaching

SDU

2026-2027 **Software Engineering Program**: Supervision of Software Engineering Semester Projects.

UTFPR

2024–2025 **Master's in Applied Computer Science**: Green Software Engineering.

2024–2025 **Bachelor's in Information Systems**: Requirements Engineering; Software Design and Modeling; Software Construction, Validation and Maintenance; Distributed Applications.

2019–2025 **Bachelor's in Computer Science**: Educational Robotics; Distributed Applications.

2023 **Bachelor's in Computer Science**: Software Engineering, Systems Analysis.

2022 **Bachelor's in Computer Science**: Software Engineering; Systems Analysis; Distributed Applications.

2021 **Bachelor's in Computer Science**: Distributed Applications; Software Engineering; Introduction to Programming Languages.

2020 **Bachelor's in Computer Science**: Computer Architecture; Software Engineering.

2019 **Bachelor's in Computer Science**: Computer Architecture; Distributed Applications.

2018 **Bachelor's in Computer Science**: Computing 1 (Algorithms); Computer Networks; Distributed Applications.

2017 **Bachelor's in Computer Science**: Computer Networks; Software Engineering; Data Structures.

2013 **Bachelor's in Computer Science**: Databases; Computing I (Algorithms); Instrumental Informatics; Frameworks for Data Storage and Persistence.

FAF

2012 **Technology in Internet Systems**: Databases II; Programming Language III (Java for Web).

2011 **Technology in Internet Systems**: Server Administration, Basic Internet Environment Concepts; E-commerce and Marketing; Object-Oriented Programming.

2010 **Technology in Internet Systems**: Server Administration for the Internet; Databases II; Multimedia Systems for the Internet Environment.

2009 **Technology in Internet Systems**: Databases II; Programming Language II (HTML/PHP); Basic Informatics Concepts.

Unisep

2011 **Bachelor's in Information Systems**: Computer Networks II; Computer Networks I; Distributed Systems; Web Infrastructure.

2010 **Bachelor's in Information Systems**: Distributed Systems.

2009 **Bachelor's in Information Systems**: Complementary Curricular Activities II (Basic Linux).

UNOESC

2012 **Bachelor's in Information Systems**: Theory of Computation; Final Graduation Project II; Databases I; Advanced Topics in Databases.

Publications

Journal Articles

2025 HAMER, E.; ALBONICO, M.; MALAVOLTA, I. **Resource Utilization of 2D SLAM Algorithms in ROS-Based Systems: an Empirical Evaluation**. *Journal of the Brazilian Computer Society*, 2025.

ALBONICO, M.; CANIZZA, M. B.; WORTMANN, Andreas. **Energy Efficiency in ROS Communication: A Comparison Across Programming Languages and Workloads**. *Frontiers In Robotics And AI*, vol. 12, p. 1, 2025.

SOLIMAN, M.; ALBONICO, M.; MALAVOLTA, I.; WORTMANN, Andreas. **Mining Software Repositories for Software Architecture – a Systematic Mapping Study**. *Information And Software Technology*, vol. 1, p. 1, 2025.

2023 ALBONICO, M.; DORDEVIC, M.; HAMER, E.; MALAVOLTA, I. **Software engineering research on the Robot Operating System: A systematic mapping study**. *Journal of Systems and Software*, vol. 197, p. 111574, 2023.

DORDEVIC, M.; ALBONICO, M.; LEWIS, G. A.; MALAVOLTA, I.; LAGO, P. **Computation Offloading for Ground Robotic Systems Communicating over WiFi – An Empirical Exploration on Performance and Energy Trade-offs**. *Empirical Software Engineering (EMSE)*, 2023.

2020 VARELA, P.; ALBONICO, M.; JUSTINO, E.; ASSIS, J. **Authorship Attribution in Latin Languages using Stylometry**. *IEEE Latin America Transactions*, vol. 18, p. 729-735, 2020.

Book Chapters

2018 ALBONICO, MICHEL; MOTTU, JEAN-MARIE; SUNYÉ, GERSON; ALVARES, FREDERICO. **Controlling Cloud-Based Systems for Elasticity Test Reproduction**. *Communications in Computer and Information Science*, Springer International Publishing, vol. 864, p. 200-222, 2018.

Conference Papers

2026 BORGES, M. F.; ALBONICO, M.; FABRI, J. A. **On the Architectural Evolution of ROS 2: Toward Greener Robotics Software Design**. *European Conference on Software Architecture (ECSA)*, p. 3–10, 2026.

ALBONICO, M.; SOUZA, D. **Silent ROS Community: Let's Start Discussing the Impact of AI Adoption on Public Knowledge Sharing**. *Brazilian Symposium on Software Engineering (SBES), Insightful Ideas and Emerging Results (IIER) Track*, 2026. (Under publication).

ALBONICO, M.; MOREIRA, N. M. **Whispers and Watts: An Empirical Study of the Energy Consumption of ROS 2 Topic-Based Communication**. *International Symposium on Empirical Software Engineering and Measurement (ESEM)*, 2026. (Under publication).

APSAN, R.; STOICO, V.; ALBONICO, M.; DHAR, R.; VAIDHYANATHAN, K.; MALAVOLTA, I. **Generating Energy-Efficient Code via Large-Language Models–Where Are We Now?**. *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (ICSE)*, 2026.

ALBONICO, M.; WORTMANN, A.; MALAVOLTA, I. **Tuning ROS 2 for Energy-Efficient Navigation: Empirical Insights from Costmap 2D Configurations**. *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

CANNIZZA, M. B.; ALBONICO, M. **A Curated List of Open-source Software-only Energy Efficiency Measurement Tools: A GitHub Mining Study**. *GreenArch Workshop at the IEEE International Conference on Software Architecture (ICSA)*, 2026.

ALBONICO, M.; CALDAS, R. D.; KEEN, H. E. **Proceedings of the 8th International Workshop on Robotics Software Engineering, co-located with the 2026 IEEE International Conference on Robotics and Automation (ICRA 2026)**. *CEUR Workshop Proceedings*, 2026.

2024 ALBONICO, M.; VARELA, PAULO J.; ROHLING, ADAIR; WORTMANN, Andreas. **Energy Efficiency of ROS Nodes in Different Languages: Publisher-Subscriber Case Studies**. In: *International Workshop on Robotics Software Engineering (RoSE)*, 2024, Lisboa, Portugal.

2023 ALBONICO, MICHEL; VARELA, PAULO JÚNIOR. **A Report on the Use of ChatGPT in Software Engineering and Systems Analysis Courses**. In: *SBES 2023: XXXVII Brazilian Symposium on Software Engineering*, Campo Grande, Brazil. Proceedings of the XXXVII Brazilian Symposium on Software Engineering, ACM, p. 303.

2022 MAYER, ROBERTO; VARELA, PAULO; ALBONICO, MICHEL; ROHLING, ADAIR; STEFFEN, VILMAR. **Experiências de um Jogo Educacional Digital para auxiliar no Processo de Ensino-Aprendizagem de Transformações Químicas para o Ensino Médio**. In: *Workshop de Informática na Escola*, 2022, Brasil. Anais do XXVIII Workshop de Informática na Escola (WIE 2022), p. 59.

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